

From the Dengue Exercise to ARGO

One regression, grown step by step into a real forecaster

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Where we pick up

Yesterday you fit a **static** least-squares line: dengue cases from the dengue search term, trained once on 2004–2006.

Today, four small upgrades, same dengue data

Static → **dynamic** training → **more terms** → **autoregression** = **ARGO**. Then borrow from neighbors (ARGONet).

- It stays an **interpretable regression** the whole way. No black box.
- Same agent workflow: one prompt per upgrade, and you check the error each time.
- Continues Mauricio's exercise 2; the pieces are the same, the destination is ARGO.

Static vs dynamic training

Adding predictive terms

Adding autoregression

Arriving at ARGO

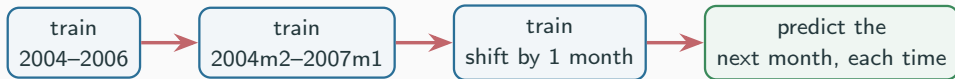
Network approaches: ARGONet

Wrap-up

Static vs dynamic training

The problem with training once

A static model freezes the search-to-disease relationship in 2004–2006. But that relationship **drifts**: behavior changes, Google changes, the disease changes.



- **Dynamic training:** slide a fixed window (36 months), refit every step, predict the next month. The model stays current.
- This is the fix Google Flu Trends lacked, and the reason it eventually drifted.

Prompt

```
Take MX_Dengue_trends.csv. Implement a sliding-window
dynamic model: for each month from 2007 on, train least
squares (Dengue CDC ~ dengue) on the previous 36 months
and predict the next month. Plot dynamic vs static vs
actual, and report the RMSE of each. Which is better?
```

Refitting on recent data alone usually beats the frozen line. That is the whole idea of data assimilation.

Adding predictive terms

One term is leaving signal on the table

You scraped several dengue-related searches. Use them all, not just dengue.

Prompt

```
Extend the dynamic model to use all four search terms as  
predictors: dengue, sintomas de dengue, mosquito, dengue  
sintomas. Recompute the RMSE and compare to the one-term  
dynamic model.
```

- More terms usually helps, but many correlated terms can **overfit** a short window.
- The standard cure is **Lasso** (ℓ_1): keep the useful terms, zero the rest. Worth adding once you have a scraped pile of queries.

Adding autoregression

The disease's own recent past is a strong predictor

Search says “something is happening now.” The last months of *cases* say “and here is the trend it is riding.” Add lagged case counts as features.

AR terms: y_{t-1} , y_{t-2} (last two months of cases).

Prompt

```
Add autoregression: include last month's and the month
before's dengue cases (AR1, AR2) as extra features in the
dynamic multi-term model. Refit the sliding window and
compare RMSE to the previous models.
```

Hint from the exercise: shift the case column, stack it on the feature matrix.

Arriving at ARGO

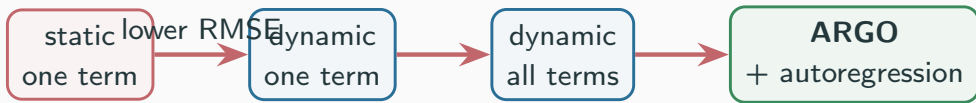
Autoregression + Google search = ARGO

Put the disease's history and the search terms in one dynamically-trained regression:

$$y_t = \mu + \sum_{k=1}^K \beta_k y_{t-k} + \sum_{j=1}^N \alpha_j x_{j,t} + \varepsilon_t.$$

- **Autoregression** (y_{t-k}) carries the momentum; **search** ($x_{j,t}$) supplies the early turn. Together they beat either alone.
- That is **ARGO** (AutoRegression with Google, Yang, Santillana & Kou, PNAS 2015). The exercise's "Google + AR2, dynamic" is ARGO in its basic form.
- The full ARGO adds an ℓ_1 **penalty** (for many terms) and a rolling two-year window. Same equation, regularized.

The error drops at every step

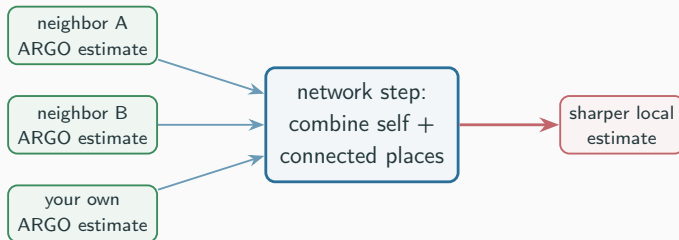


- Each upgrade is one honest, out-of-sample comparison on 2007–2011.
- Nothing here is a black box: you can read every coefficient and say what it does.

Network approaches: ARGONet

A location is not an island

ARGO uses one location's own search and history. But your **neighbors** carry signal too: what is rising next door often reaches you next (the mobility link from this morning).



- **ARGONet** (Lu, Yang, Santillana et al., *Nature Communications* 2019) adds a step that borrows strength from connected locations.
- There is a finer **county-level** version too. Same principle, more locations.

How the network step helps

- Which neighbors? The ones with the most **shared movement** (the LODS / mobility ranking from this morning), or simply the most correlated at a short lag.
- Feed a neighbor's recent activity, or its ARGO estimate, in as an extra predictor, and let the dynamic fit decide how much to trust it.
- Where local data is thin or late, a well-connected neighbor can fill the gap.

The dengue CSV is one country, so this is a concept here; on the state or county panels it is a real, measurable gain.

Wrap-up

What you built

- A single regression grown into **ARGO**: dynamic training + selected search terms + autoregression, validated out of sample at each step.
- A network idea (**ARGONet**) that borrows signal from connected neighbors.
- All of it **interpretable**, and all of it driven by prompts you can rerun for your own disease and place.

This is the operational machinery Candice runs weekly for Georgia flu. You just built the small version.

- **Next (Part 2, M. Santillana):** beyond flu monitoring in rich nations, emerging outbreaks, and weather as a modulator.
- **This afternoon (3:30):** the COVID exercise, detecting exponential growth in digital traces.
- **Day 3:** bring your own problem, and point ARGO at it.

From one search term and a straight line, to ARGO, in an afternoon.

- Exercise continues Mauricio Santillana's dengue exercise 2 (MIGHTE-1ab/SISMID25); data `MX_Dengue_trends.csv`.
- **ARGO**: Yang, Santillana & Kou, *Accurate estimation of influenza epidemics using Google search data via ARGO*, PNAS 2015.
- **ARGONet**: Lu, Yang, . . . , Santillana, *Improved state-level influenza nowcasting in the United States leveraging Internet-based data and network approaches*, Nature Communications 2019.
- Dynamic training as data assimilation; Lasso for term selection with many queries.

Questions?

Repo: `github.com/Shihao-Yang/sismid2026`